

Injaaz App

Facilities & operations platform — architecture and modules
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1. Purpose

Injaaz is a production-ready Flask application for facilities and operations: digital site-visit and inspection forms, multi-stage review workflows, reporting, document generation (including HR forms as PDF and DOCX), and centralized administration. The codebase is organized as a **modular monolith**—one deployable app with feature blueprints sharing authentication, database, and services.

2. Technical foundation

Area	Technology
Runtime	Python, Flask
API auth	JWT (Flask-JWT-Extended)
Database	SQLAlchemy; SQLite (dev), PostgreSQL (production)
Frontend	HTML, CSS, JavaScript; PWA assets; Capacitor for mobile shells
Files / media	Local uploads; Cloudinary in production where configured
Email	Mailjet (e.g. MMR scheduled reports)
Production	wsgi.py; documented for Render and similar hosts

3. Core platform

- Authentication (**/api/auth**) — login, registration, access and refresh tokens.
- Dashboard and main UI — gated by role and module permissions.
- Admin (**/api/admin**) — users, access flags, device and operational admin tools.
- Workflow (**/api/workflow**) — submission pipelines, pending reviews, history.
- Reports API (**/api/reports**) — on-demand reporting where implemented.

4. Operational modules

Module	Typical URL	Role
HVAC / MEP	/hvac-mep	HVAC & MEP forms, generators, reporting
Civil	/civil	Civil works forms and reports
Cleaning	/cleaning	Cleaning services forms and reports
Inspection	/inspection	Cross-trade inspection workflows
HR	/hr	HR forms, HR→GM approval, PDF/DOCX output
Procurement	/procurement	Procurement workflows
MMR	/admin/mmr	CAFM analytics, Excel reports, scheduled email (Dubai TZ)
Business Development	/bd, /admin/bd	BD flows and admin entry
DocHub	/api/docs, /dochub	Authorized document hub

5. How the pieces fit together

Users authenticate and open the dashboard; permissions control admin, workflow, and module routes. Field modules capture structured data and attachments. Workflow advances records through supervisors and reviewers. HR submissions follow **employee submit** → **HR review and sign** → **General Manager final approval**, with printable HTML and downloadable PDF/DOCX. MMR ingests CAFM exports and applies business rules for chargeable analysis. Configuration is centralized in **config.py** and environment variables (database, Redis, secrets, mail, schedules).

6. Repository documentation

For setup and deployment: **README.md**, **SETUP.md**, **QUICK_START.md**. For structure and flows: **PROJECT_STRUCTURE.md**, **PROJECT_FLOW.md**. For architecture detail: **docs/APPLICATION_OVERVIEW.md**. Production notes: **CLOUD_ONLY_SETUP.md**, **RENDER_DEPLOYMENT_PHASES.md**.

This PDF is generated by `scripts/generate_project_overview_pdf.py` and summarizes the Injaaz application for stakeholders and onboarding. Update the script when major modules or URLs change.